

UCI Complex Analysis Qualifying Solutions

Jinghao Chen
Department of Mathematics, UC Irvine

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Problem 1.

Suppose f is an entire holomorphic function and

$$\iint_{\mathbb{R}^2} |f(x + iy)|^2 dx dy < \infty.$$

Prove that $f \equiv 0$.

Proof:

$$\int_{\mathbb{C}} |f(z)|^2 dA(z) = \iint_{\mathbb{R}} |f(x + iy)|^2 dx dy < \infty$$

so say $\exists M < \infty$, s.t. $\int_{\mathbb{C}} |f(z)|^2 dA(z) = M < \infty$ bounded.

$\forall z \in \mathbb{C}$, by **Mean Value Principle**¹,

$$f^2(z) = \frac{1}{\pi R^2} \int_{D(z, R)} f^2(w) dA(w)$$

holds for any $R > 0$ as f is entire.

So

$$|f(z)|^2 \leq \frac{1}{\pi R^2} \int_{D(z, R)} |f(w)|^2 dA(w) \leq \frac{1}{\pi R^2} \int_{\mathbb{C}} |f(w)|^2 dA(w) = \frac{M}{\pi R^2}$$

because radius R is arbitrary, so $|f(z)|^2 \leq \frac{M}{\pi R^2} \rightarrow 0$ as $R \rightarrow \infty$.

Hence $|f(z)|^2 \equiv 0 \Rightarrow |f(z)| \equiv 0 \Rightarrow f(z) \equiv 0 \forall z \in \mathbb{C}$. $\square$

¹MVP Ver.2, 2 dimensional case

Problem 2.

Suppose f is a holomorphic function on the disk $D(0, R)$ about the origin with radius $R > 1$, and that $|f(z)| < 1$ for $|z| = 1$. Prove that there exists a unique point z_0 in the disk such that $f(z_0) = z_0$.

Proof:

Let $g(z) := f(z) - z$ and $h(z) := -z$, then for any $z \in \partial D(0, 1)$, i.e. $|z| = 1$,

$$|g(z) - h(z)| = |f(z)| < 1 = |z| = |h(z)|,$$

so by **Rouche Theorem**,

$$\#\mathbb{Z}_{D(0,1)}(g) = \#\mathbb{Z}_{D(0,1)}(h) = 1,$$

i.e. $g = f - z$ has an unique root in the unit disk, which means that unique root is exactly its unique fixed point s.t. $f(z_0) = z_0$. $\square$

Problem 3.

Consider the function $f(z) = z \cos \frac{\pi}{z}$. Find $\int_{\gamma} f(z) dz$, where the closed curve γ is given by $\gamma(t) = e^{2it} + \frac{1}{2}e^{3it}$, $t \in [0, 2\pi]$.

Sol:

$f(z)$ has one singularity 0. Let's start with computing the index of function f around 0:

$$\begin{aligned} \text{Ind}_f(0) &= \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z} dz \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{1}{\gamma(t)} \gamma'(t) dt \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{2ie^{2it} + \frac{3i}{2}e^{3it}}{e^{2it} + \frac{1}{2}e^{3it}} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{4 + 3e^{it}}{2 + e^{it}} dt \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{4 + 3e^{it}}{2e^{it} + e^{2it}} ie^{it} dt \\ &= \frac{1}{2\pi i} \int_{|z|=1} \frac{4 + 3z}{2z + z^2} dz \end{aligned}$$

the only singularity of $\frac{4+3z}{2z+z^2}$ in $D(0, 1)$ is 0, a simple pole. So

$$\text{Res}_{\frac{4+3z}{2z+z^2}}(0) = \left. \frac{4+3z}{2+z} \right|_{z=0} = 2,$$

by **Residue Theorem**,

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{4+3z}{2z+z^2} dz = \frac{1}{2\pi i} \cdot 2\pi i \cdot 2 = 2,$$

so $\text{Ind}_f(0) = 2$. By expansion:

$$\cos \frac{\pi}{2} = 1 - \frac{1}{2} \cdot \left(\frac{\pi}{2}\right)^2 + \frac{1}{24} \cdot \left(\frac{\pi}{z}\right)^4 + \dots$$

so

$$f(z) = z \cos \frac{\pi}{z} = z - \frac{\pi^2}{2} \cdot \frac{1}{z} + \frac{\pi^4}{24} \cdot \frac{1}{z^3} + \dots$$

by the uniqueness of Laurent expansion,

$$\text{Res}_f(0) = a_{-1} = -\frac{\pi^2}{2}$$

so by **Residue Theorem**,

$$\begin{aligned}\int_{\gamma} f(z)dz &= 2\pi i \cdot \text{Ind}_f(0) \cdot \text{Res}_f(0) \\ &= 2\pi i \cdot 2 \cdot -\frac{\pi^2}{2} \\ &= -2\pi^3 i\end{aligned}$$

Problem 4.

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be a non-constant nowhere vanishing entire holomorphic function. Let $U = \{z \in \mathbb{C} : |f(z)| < 1\}$. Show that U is unbounded.

Proof:

Assume U is bounded. Because f is entire and $f(z) \neq 0$ for any $z \in \mathbb{C}$, i.e. nowhere vanishing. So let $g(z) := \frac{1}{f(z)}$ is also an entire holomorphic function, and $U = \{z \in \mathbb{C} : |f(z)| < 1\} = \{z \in \mathbb{C} : |g(z)| > 1\}$ is bounded, which means exist bounded closed set V s.t. $U \subseteq V$. So

$$\max_{z \in V} |g(z)| = \max_{z \in \partial V} |g(z)| = M < \infty$$

for some $M > 0$ by **Maximum Modulus Theorem**.

On the other hand, for any $z \in \mathbb{C} \setminus V$, because $z \notin U$, so $|g(z)| \leq M$ on $\mathbb{C}$ uniformly bounded. By **Liouville Theorem**, g is a constant, so $f = \frac{1}{g}$ is also a constant, contradict to f non-constant! Hence, U should be unbounded. $\square$

Problem 5.

Let f be a function holomorphic in the open unit disc, bounded by 1 in absolute value, and vanishing at the origin. Show that the sum

$$\sum_{n=1}^{\infty} f(z^n)$$

converges uniformly on compact subsets of the unit disc to a holomorphic function.

Proof:

Because $|f(z)| \leq 1$ on $\mathbb{D} = D(0, 1)$, so

$$f : \mathbb{D} \rightarrow \mathbb{D}, \quad 0 \mapsto 0,$$

so by **Schwarz Lemma**, $|f(z)| \leq |z|$ on $\mathbb{D}$.

For any compact subset Ω of $\mathbb{D}$, $\exists R$ where $0 < R < 1$, s.t. $\Omega \subseteq D(0, R) \subseteq \mathbb{D}$. So on Ω :

$$\left| \sum_{n=1}^{\infty} f(z^n) \right| \leq \sum_{n=1}^{\infty} |f(z^n)| \leq \sum_{n=1}^{\infty} |z^n| = \sum_{n=1}^{\infty} |z|^n \leq \sum_{n=1}^{\infty} R^n < \infty$$

as $R < 1$.

Hence $\sum_{n=1}^{\infty} f(z^n)$ converges uniformly and absolutely on any compact subsets of $\mathbb{D}$. $\square$

Additional Notes: The uniform limit of holomorphic functions is holomorphic. $f(z^n)$ holomorphic, so the partial sum $\sum_{n=1}^N f(z^n)$ is also holomorphic, so $\sum_{n=1}^{\infty} f(z^n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N f(z^n)$ converges to a holomorphic function.

Problem 6.

Prove or disprove:

a) There exists a continuous real valued non-constant function u on the square $\Pi = \{0 \leq \operatorname{Re} z \leq 1, 0 \leq \operatorname{Im} z \leq 1\}$, harmonic in its interior, such that $u(z) = 0$ for each $z \in [0, 1]$.

b) There exists a continuous complex valued non-constant function f on the square $\Pi = \{0 \leq \operatorname{Re} z \leq 1, 0 \leq \operatorname{Im} z \leq 1\}$, holomorphic in its interior, such that $f(z) = 0$ for each $z \in [0, 1]$.

Sol:

a) TRUE.

Consider $u(z) = u(x + iy) = y$, u is non-constant and $\Delta u \equiv 0$ so harmonic. When $z \in [0, 1] \subseteq \mathbb{R}$ then $u(z) = y = 0$.

b) FALSE.

Assume such holomorphic $f(z)$ exists. Because $f(z) = 0 \in \mathbb{R}$ on $[0, 1]$, by **Schwarz Reflexive Principle**, $f(z)$ can be extended to be holomorphic on $\{0 \leq \operatorname{Re} z \leq 1, -1 \leq \operatorname{Im} z \leq 1\}$ by

$$\tilde{f}(z) := \begin{cases} f(z) & z \in \{0 < \operatorname{Re} z < 1, 0 < \operatorname{Im} z < 1\} \\ 0 & z \in (0, 1) \subseteq \mathbb{R} \\ \overline{f(\bar{z})} & z \in \{0 < \operatorname{Re} z < 1, -1 < \operatorname{Im} z < 0\} \end{cases}$$

However, $\tilde{f}(z) = 0$ on $(0, 1)$ which is an accumulative set, by **Uniqueness Principle**², $\tilde{f}(z) \equiv 0 \Rightarrow f(z) \equiv 0$, contradict with $f(z)$ non-constant! $\square$

²also known as **Identity Principle**, **Isolated Zeros Theorem**, etc.

Problem 7.

Prove or disprove: there exists a sequence $\{f_n\}$ of holomorphic functions on the unit disc $D(0, 1)$ such that $f_n \rightarrow \bar{z}^3$ uniformly on compact subsets of $D(0, 1)$.

Claim: the statement is FALSE.

Assume such holomorphic sequence exists, while $\partial D(0, \frac{1}{2})$ is a compact subset of $D(0, 1)$, so on it:

$$f_n \xrightarrow{\text{uniformly}} \bar{z}^3 \Rightarrow z^2 f_n \xrightarrow{\text{uniformly}} z^2 \bar{z}^3$$
$$z^2 \bar{z}^3 = z^2 \bar{z}^2 \bar{z} = |z|^2 \bar{z} = |z|^2 \cdot \frac{|z|}{z} = |z|^3 \cdot \frac{1}{z} = \frac{1}{8z}$$

On the other hand,

$$\int_{\partial D(0, \frac{1}{2})} z^2 f_n dz = 0, \forall n \in \mathbb{N}$$

by **Cauchy Integral Theorem** as $z^2 f_n$ holomorphic. So

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \int_{\partial D(0, \frac{1}{2})} z^2 f_n dz \\ &= \int_{\partial D(0, \frac{1}{2})} \lim_{n \rightarrow \infty} z^2 f_n dz \\ &= \int_{\partial D(0, \frac{1}{2})} \frac{1}{8z} dz \\ &= \frac{\pi i}{4} \neq 0 \end{aligned}$$

contradict!

So such sequence doesn't exist. $\square$

Problem 8.

Fix $0 < \gamma < 1$ and suppose f is a holomorphic function on the unit disc satisfying

$$|f(z) - f(w)| \leq C_1 |z - w|^\gamma$$

for some constant C_1 . Show that there exists a constant $C_2 > 0$ such that

$$|f'(z)| \leq C_2 (1 - |z|)^{\gamma-1}.$$

Proof:

For any fixed $z \in \mathbb{D} = D(0, 1)$, define function of w :

$$g(w) := \frac{f(z) - f(w)}{z - w}$$

Since f is holomorphic on $\mathbb{D}$, g is holomorphic on $D(z, R) \setminus \{z\} \subseteq \mathbb{D}$ for any R where $0 < R \leq 1 - |z|$. Since

$$\lim_{w \rightarrow z} g(w) = \lim_{w \rightarrow z} \frac{f(z) - f(w)}{z - w} = f'(z) < \infty,$$

by **Riemann Removable Singularity Theorem**, z is removable for g , so g is actually holomorphic on $D(z, R)$ with extended value $g(z) := f'(z)$.

On the other hand, since $|f(z) - f(w)| \leq C_1 |z - w|^\gamma$, so

$$|g(w)| = \left| \frac{f(z) - f(w)}{z - w} \right| \leq C_1 |z - w|^{\gamma-1}$$

for all $w \in D(z, R)$. So for $w \in \partial D(z, R)$, $|g(w)| \leq C_1 R^{\gamma-1}$. Now, by **Maximum Modulus Principle**,

$$|f'(z)| = |g(z)| \leq \max_{w \in \partial D(z, R)} |g(w)| \leq C_1 R^{\gamma-1}.$$

Because g is holomorphic in $D(z, R)$ for any R s.t. $D(z, R) \subseteq \mathbb{D}$, so the inequality above should hold for any satisfied R . Recall that $0 < R \leq 1 - |z|$, so

$$|f'(z)| \leq C_1 (1 - |z|)^{\gamma-1},$$

hence $C_2 = C_1$ is exactly what we desired. $\square$

The complete file of all Qual solutions can be requested in person only for some policy reasons.